



EAST DELTA UNIVERSITY

Title

PYTHON PROJECT PROPOSAL

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Arshiana Shamir

Lecturer, Department of Computer Science & Engineering

Course

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Prepared By

Ahmed Noorim

Student ID: **212000512**

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Title

Detecting Bengali Alphabets from handwriting using
Scratch CNN and Implementing it in a Flask Web App

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Abstract

This project presents a deep learning-based approach for the classification of Bengali alphabets using a scratch Convolutional Neural Network (CNN). The model achieves an accuracy of 91% in recognizing individual Bengali characters from images, demonstrating its effectiveness in character detection. Implemented as a web application using Flask, the system allows users to input images and receive real-time classification results. This work addresses the critical need for efficient detection of the Bengali alphabet, particularly in handwritten forms, and lays the groundwork for future advances in optical character recognition and accessibility tools for the Bengali-speaking community.

Introduction

In many applications such as educational tools, automated transcription services, social media, there is a need to recognize and classify Bengali alphabets. Bengali is one of the biggest languages in the world with more than 270 million speakers, yet it still lags in digital representation and processing. Handwritten and printed text recognition are two distinct tasks, and there is a need to bridge this gap for efficient recognition of Bengali alphabet characters in order to improve communication and access to information in the digital world.

Deep learning is a subset of ML that uses large datasets and Neural Networks with multiple layers to deduce. This approach has revolutionized problems of image recognition which teach models to learn complex patterns and features directly from raw data. Moreover, at present, Convolutional Neural Networks (CNNs) have shown to be extremely effective for image classification tasks since they are so well suited to capture spatial hierarchies of images. In this project, deep learning algorithms are used to develop a scratch CNN model to classify Bengali alphabets from handwriting images.

Background

In the recent past, with the increase in the availability of handwriting data, advancements in optical character recognition (OCR) systems for languages like English have been largely relegated to deep learning models designed to recognize specific language scripts. It has been shown that CNN models provide good accuracy for character recognition compared to traditional ML methods. The superiority of deep learning models also depends on the size of the dataset. Variances in handwriting styles ensure that the model can generalize effectively. The challenges of accurately detecting handwritten characters in Bengali highlight the need for better-equipped models and more comprehensive datasets. These are crucial to generalize across different variations in a writer's handwriting style and conditions.

Problem Statement

Despite improvements in character recognition in other languages, there still persists a huge gap in the detection of handwritten Bengali alphabets. To date, most existing systems are primarily dedicated to printed text while a considerable portion of handwritten data remains unaddressed. Another significant obstacle is the development of applications requiring large amounts of handwritten documents. By addressing these challenges, this work significantly advances the state of the art in Bengali alphabet detection from handwritten text by proposing a scratch CNN model and a larger dataset while implementing it in a web app that will be useful to Bengali speakers.